Resources for standardizing and sharing survey and monitoring data

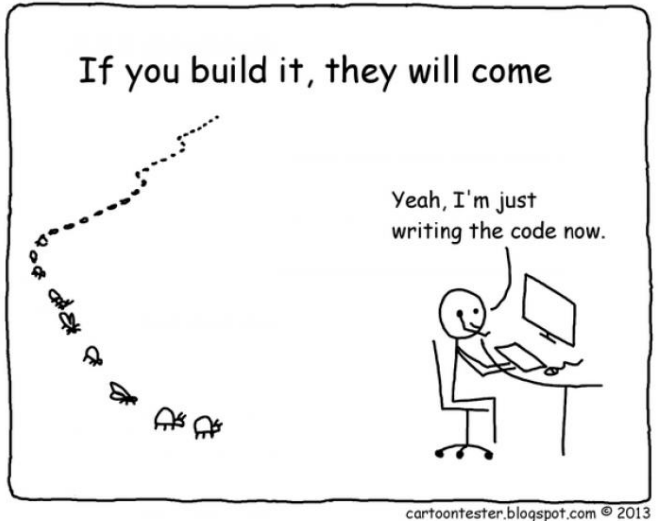


22 October 2026

Kate Ingenloff | Scientific Officer

If you build it...

- **Standards:** Event core (DwC-A)
- **Standards:** Humboldt extension
- **Guidance & support:** Guides
- **Guidance & support:** Outreach, training
- **Standards:** DwC Data Package (DwC-DP)
- **Infrastructure:** Event functionality on GBIF.org
- **Guidance & support:** Blog, updated training materials, SAM help desk

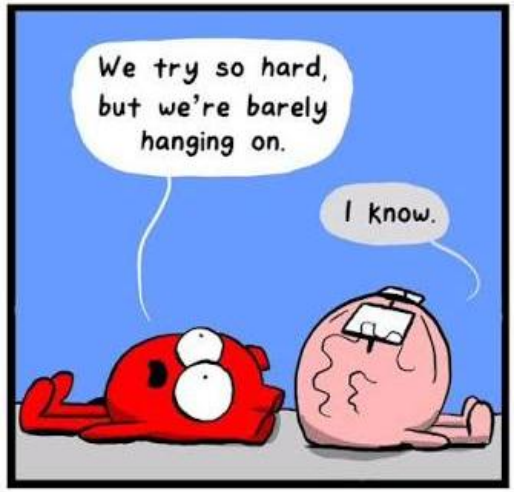


...they will come.

Impediments

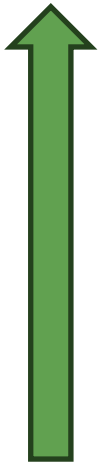
- Limited resources
- Competing priorities
- No capacity to even consider changing up the pipelines
- Infrastructure limitations





Standards: 1 DwC standard, 2 DwC data models

https://dwc.tdwg.org



→ Darwin Core Archives (DwC-A)

- You know this!
- Simpler, flattened data structure

→ Darwin Core Data Packages (DwC-DP)

- Relational data model

Getting started

- Quick Reference Guide: current terms organized for human use, with definitions, comments, and examples
- Conceptual Model: classes of information and how they are related
- Normative term list: complete metadata for terms used in the Darwin Core vocabulary, including obsolete terms
- Darwin Core Schemas: machine-readable schemas that are part of or support Darwin Core
- Usage guides:
  - Simple Darwin Core
  - Text (Darwin Core Archive)
  - Data Package (Darwin Core Data Package)
  - Darwin Core Data Package Quick Reference Guide
  - XML
  - RDF
- GitHub repository: where Darwin Core is maintained
  - List of open issues
  - List of open milestones
- Distribution files: CSV files of lists of term names

Choose only what you need for your data.

Guides



→ Good SAMaritans blog: https://data-blog.gbif.org/tags/sam/

→ NEON mapping documents (Thank you, Chandra!)

- Mapping NEON Survey Monitoring Data to the Humboldt Extension for Ecological Inventories, https://biokic.github.io/NEONTickstoHumboldt/
- Mapping NEON Bird Survey Monitoring Data to the Humboldt Extension for Ecological Inventories, https://biokic.github.io/NEONBirdstoHumboldt/

→ Comprehensive guide: https://doi.org/10.35035/doc-ynvs-eh84

- Major update anticipated mid-November

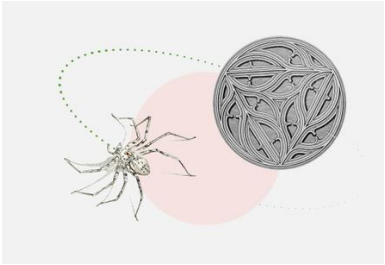
Good SAMaritans blog

https://data-blog.gbif.org/tags/sam/



Topic	Publication month
Introduction	July 2026
Scope	August 2026
Time & place	September 2026
Protocols	October 2026
Effort	November 2026
Materials	December 2026
Occurrences	January 2027
From methods section to event table	February 2027

Other resources



→ Templates

https://github.com/gbif/surveys-and-monitoring/tree/master/Templates

- Full
- Minimum recommended model

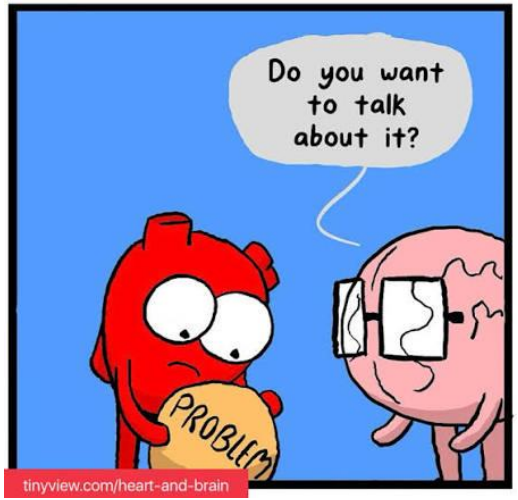
→ Training

- Biodiversity data mobilization course (https://training.gbif.org/en/data-mobilization/)

Support: *Can I just talk to a real person?!?*

Survey & monitoring (SAM) data help desk

sam@gbif.org

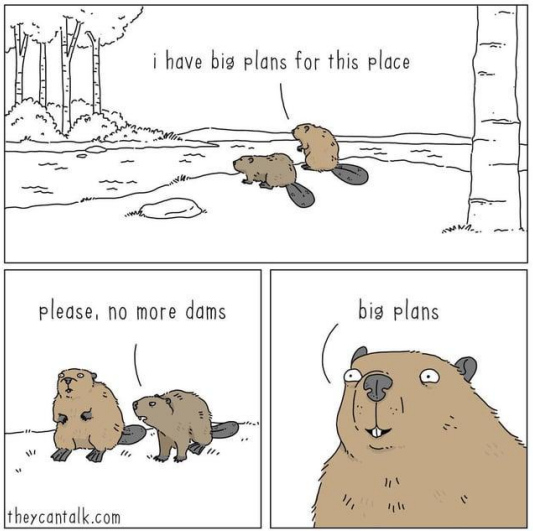


GBIF Biological surveys & Monitoring Data Hub

https://www.gbif.org/sam

Links to:

- News & events
- Guides
- GitHub
- Data publishing templates
- Help desk contact



What other resources would be helpful for you?